

Chad Galloway
CST-250 Programming in C# II
Grand Canyon University
Dec. 07, 2025
Milestone 6

Files

<https://github.com/CGalloway3/CST-250-Projects/tree/master/Milestone6>

Video

[Milestone 6 part 1](#)

[Milestone 6 Part 2](#)

FLOW CHART

/*
 * Chad Galloway
 * CST - 250 Programming in C# II
 * 12/07/2025
 * Mine Sweeper GUI App
 * Milestone 6
 * References:
 */

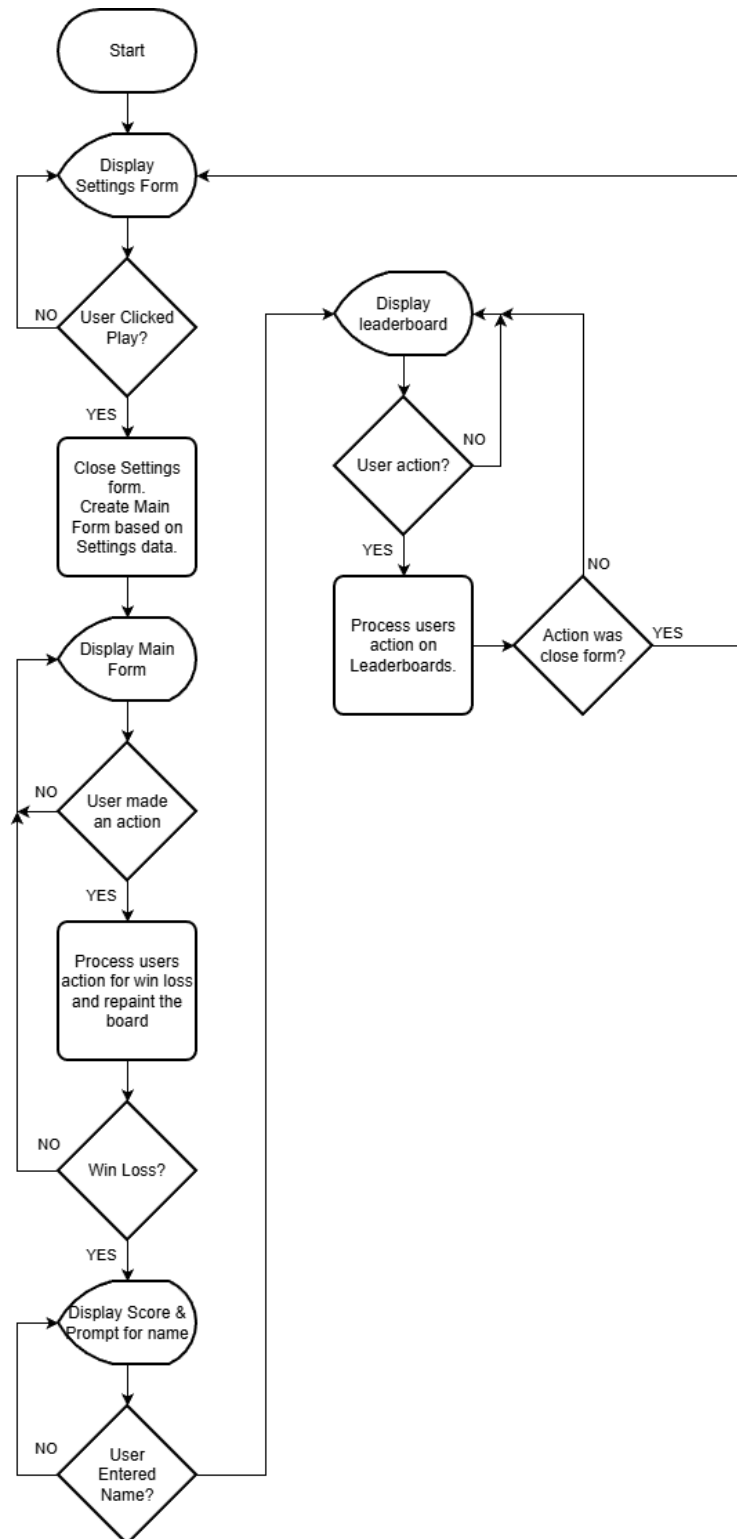


Figure 1: Flow chart of Milestone 6

UML Class Diagram

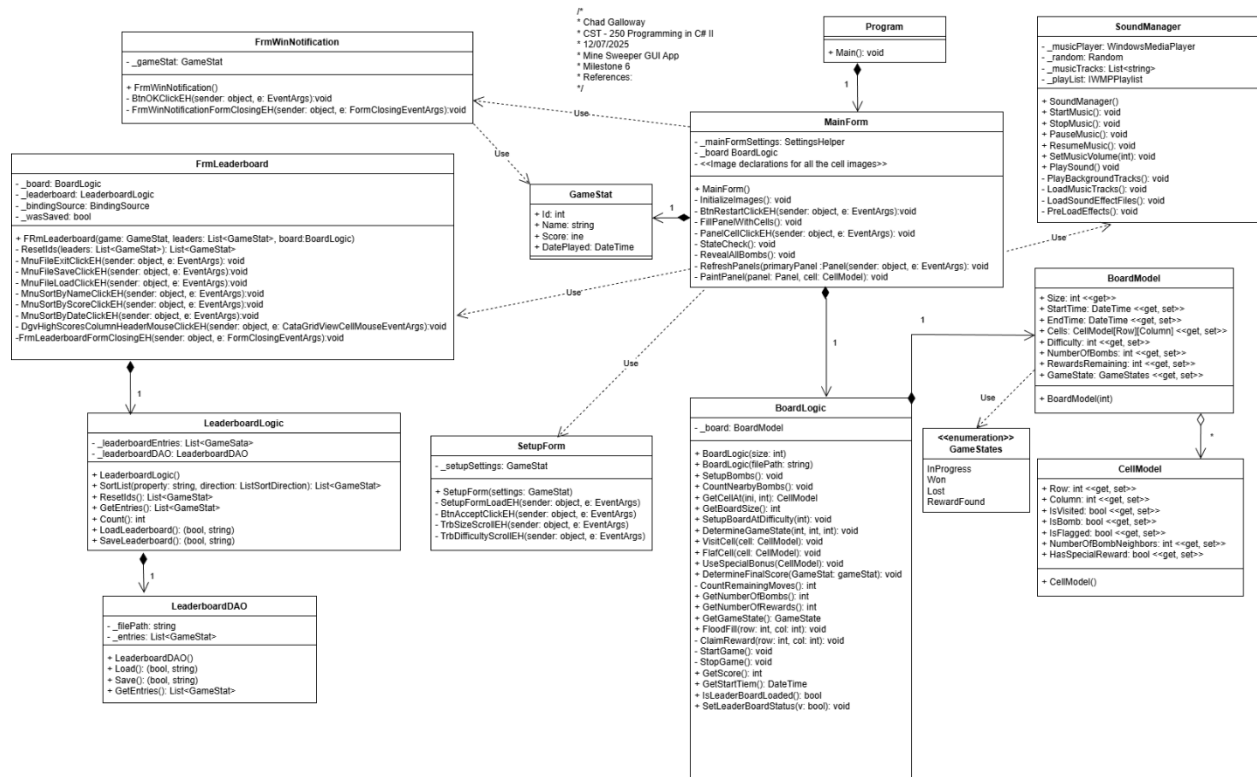


Figure 2: UML Class Diagram for Milestone 6

Screen Shots

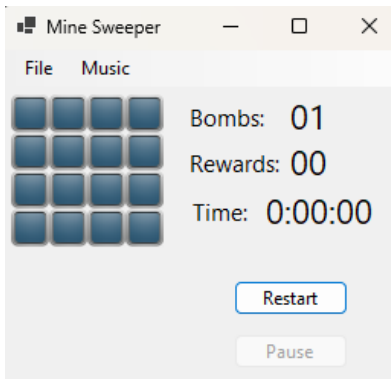


Figure 3: Screenshot of a new 4 x 4 game

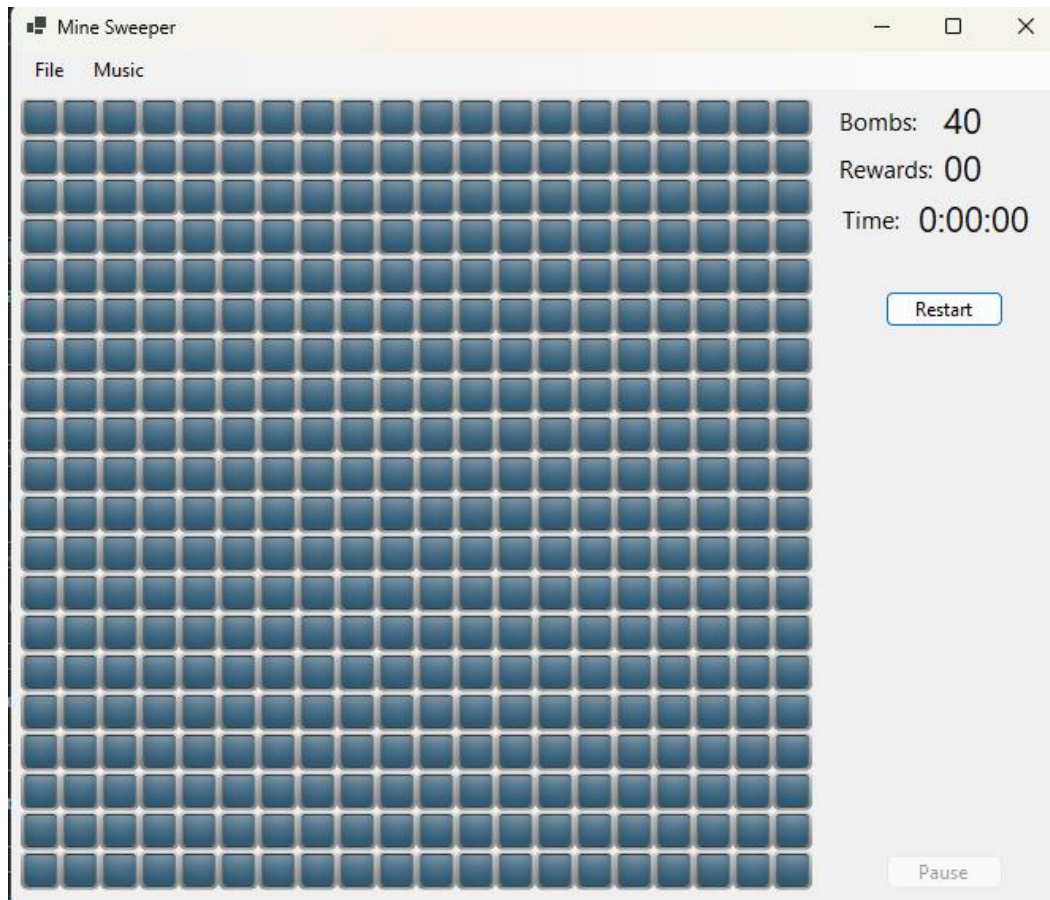


Figure 4: Screenshot of a new 20 x 20 game. You can see the menu bar added with file and music options. A pause button has also been added to the lower right of the window for pausing the game.

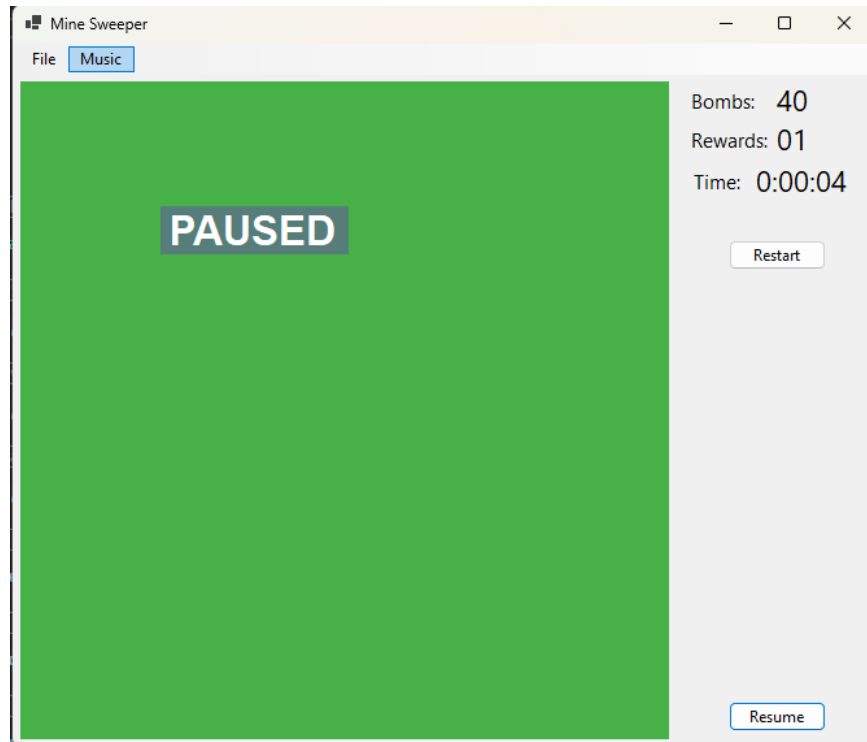


Figure 5: Screenshot of a new 20 x 20 game. While it is paused. The paused label bounces around the screen and the panel and label change colors as it moves around the screen.

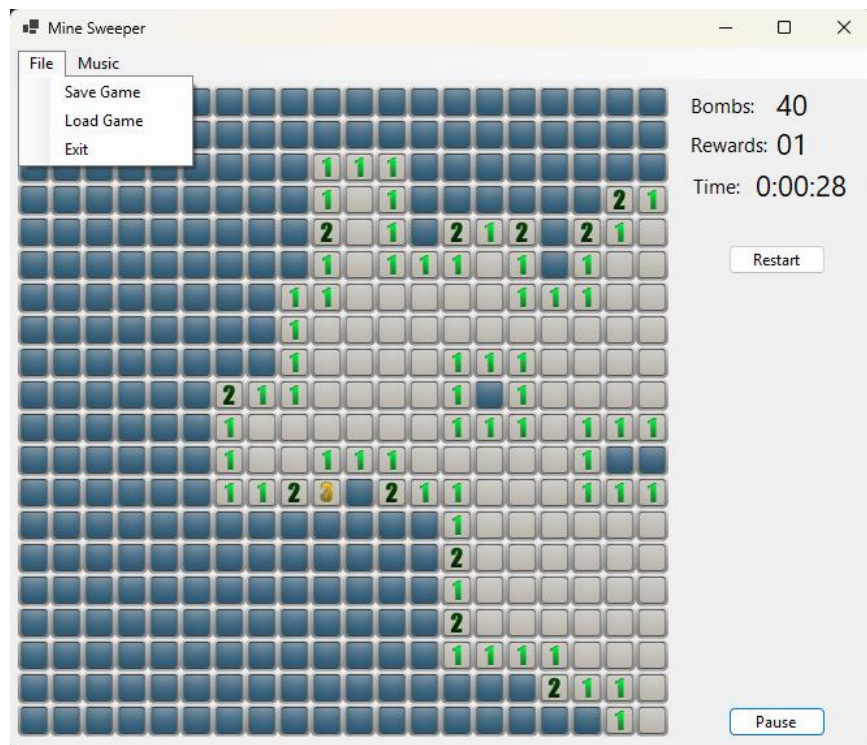


Figure 6: Screenshot of the file menu. The options are save, load, and exit. Save will write the current game state out to a file, load loads in an old file and deletes it, then exit will of course exit the form.

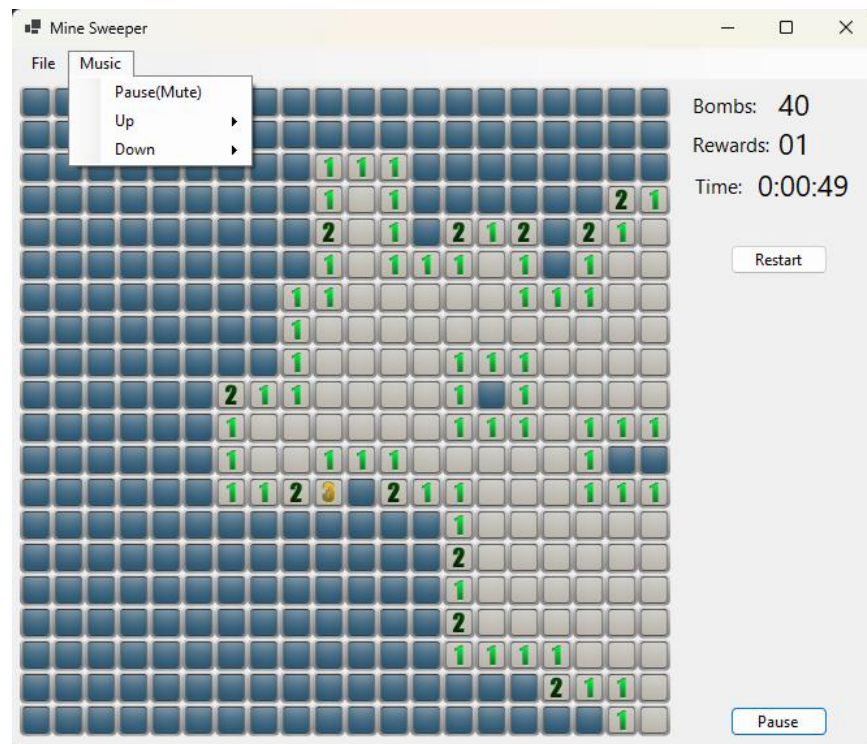


Figure 7: Screenshot of the music menu. Pause will mute and pause the music. You can resume right where you left off. One of the features I added is the ability of the user to drop whatever mp3 file they want into the music directory and it will play what they put in there.

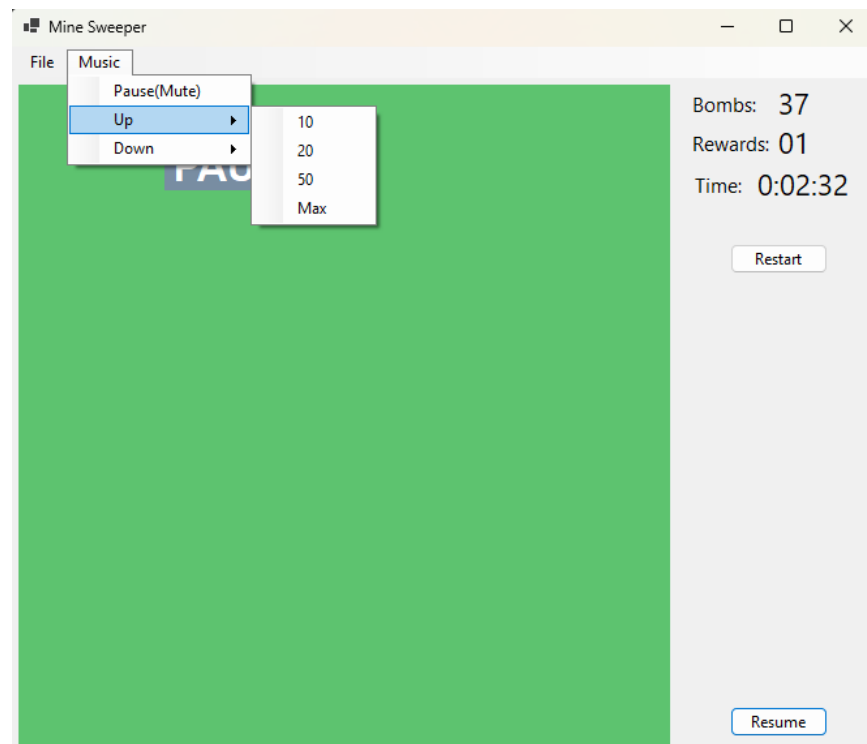


Figure 8: Screenshot of the up and down options to raise and lower volume.

This week's milestone focused on expanding the users experience while playing the minesweeper game. To achieve that goal I chose to tackle number 5 and 6 of the advanced features to add to my project. Firstly, I added in sounds and music. The events in game now have associated sounds linked to them. There is also a background music track running while the game is playing. The user can raise and lower the music's volume and pause the music if they want. The option to put their own music file into the music directory was also added into the game as a feature. Secondly, I implemented the ability to save, load, and pause a current game via the menu of as an option when closing or opening the application. I continued to use best coding practices and cleanly document all code. The separation of concerns used through out the project was maintained when adding in the new features and solid unit testing was implemented to assist in adding in the saving and loading functionality of a game.